

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER 1, 2015/2016

TCN2141 – COMPUTER NETWORKS

(All sections / Groups)

15 OCTOBER 2015
2.30 p.m. – 4.30 p.m.
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This Question paper consists of 4 pages with 7 Questions only
2. Attempt **SIX** out of **SEVEN** questions.
3. All questions carry equal marks and the distribution of the marks for each question is given.
3. Please print all your answers in the Answer Booklet provided.

QUESTION 1

- a) What is the difference between routing and forwarding?
- b) What are the differences between a classful addressing and classless addressing in IPv4? Give one example of classful routing protocol and one example of classless routing protocols.
- c) Given a host IP address with CIDR as 191.55.20.2 /23. As the network administrator, find the following information?
 - i. Find the subnet mask required.
 - ii. What is the current IP address subnet range?
 - iii. What is the subnet number for the 7th subnet?
 - iv. What is the subnet broadcast address for 4th subnet?
 - v. What are assignable addresses for the 5th subnet?

[1 + 3 + 6 = 10 Marks]

QUESTION 2

- a) When routers generate ICMP messages, to where do they send them? Along with the ICMP header at the beginning, what additional contextual information do routers include in the messages?
- b) A router receives an IP packet with source IP address 145.45.3.3 and destination IP address 201.28.4.7. The router cannot find the destination IP address in its routing table. What ICMPv4 message should be sent?
- c) List three transition strategies to move from IPv4 to IPv6. Explain the difference between tunneling and dual stack strategies during the transition period. When is each strategy used?
- d) What is the difference between a point-to-point network and a peer-to-peer network?

[2 + 2 + 3 + 3 = 10 Marks]

QUESTION 3

- a) Discuss four (4) advantages for using UDP over using TCP. State at least two applications that use these different advantages of UDP.
- b) Consider sending a 6000-byte datagram into a link that has an MTU of 1500 bytes. Suppose the original datagram is stamped with the identification number 55743 as shown in table below (Original IP datagram). How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? Use IP fragments table below to give your answer.

Continued.....

Original IP Datagram					
Sequence	Identifier	Total Length	DF	MF	Offset
0	55743	6000	0	0	0

IP Fragments (Ethernet)					
Sequence	Identifier	Total Length	DF	MF	Offset

[5+ 5 =10 Marks]

QUESTION 4

- a) What is the difference between a group-shared tree and a source-based tree in the context of multicast routing?
- b) Abbreviate the following IPv6 addresses:
- 0000:0000:0F53:6382:AB00: 67DB:BB27:7332
 - 2819:00AF:0000:0000:0000:0035:0CB2:B271
 - 140:1ABC: 419A:A000:0000:0000:0000:0000
- c) The following is a dump of a UDP header in hexadecimal form:

06 32 00 0D 00 1C E2 17

What is the :

- Source port number
- Destination port number
- Total length of the UDP
- Length of the data
- Considering that an IP frame can have a maximum total length of 65 535 bytes, what is the maximum length of the data in a UDP frame?

[2 +3 +5 = 10 Marks]

Continued.....

QUESTION 5

- a) When NAT is used, what fields in the IP and TCP headers are changed?
- b) Why is SMTP not used for transferring e-mail messages from the recipient's mail server to the recipient's personal computer?
- c) What is the difference between IP and MPLS routing?
- d) Why do you think DNS uses UDP, instead of TCP, for its query and response messages?
- e) Suppose you are sending an email from your Hotmail account to your friend, who reads his/her e-mail from his/her mail server using IMAP. Briefly describe how your email travels from your host to your friend's host. Also, what are the application-layer protocols involved?

[2+2+2+2+2 = 10 Marks]

QUESTION 6

- a) What is the maximum size of the data field for the TCP packet?
- b) Why do OSPF messages propagate faster than RIP messages?
- c) A network is using RIP for routing table update. RIP updates routing information periodically. There are N nodes in this network, and each node has B neighbor. Estimate the number of message exchanges per second.
- d) Briefly describe TCP congestion control using slow start approach.

[2+2+2+4 = 10 Marks]

QUESTION 7

- a) What is the Address Resolution Protocol (ARP) and how does it work?
- b) Briefly explain the auto-configuration of hosts feature in IPv6 addressing.
- c) What is difference between persistent and non-persistent HTTP?

[2+4+4 = 10 Marks]

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